

Day – 3 Config the RIP theory.

Task – 1 Setting up network topology with three routers.

- Network Scheme & Subnet Addressing :-
- Router 1 ↔ Router 2 Link: 10.0.0.0/30
 - o Router 1 Serial0/1/0: 10.0.0.1
 - o Router 2 Serial0/1/0: 10.0.0.2
- Router 2 ↔ Router 3 Link: 10.0.0.4/30
 - o Router 2 Serial0/1/1: 10.0.0.5
 - o Router 3 Serial0/1/0: 10.0.0.6
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- LAN Subnets:
 - o Router 1 GigabitEthernet0/0: 192.168.1.1/24 (Client LAN 1)
 - o Router 3 GigabitEthernet0/0: 192.168.2.1/24 (Client LAN 2)
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- 1.Place Devices & Install HWIC Modules:Hardware Setup.
- Drag three 2911 or 1841 Routers into the workspace (Router1, Router2, Router3).
- Click each router, go to the Physical tab, and toggle the Power Switch to OFF.
- Drag the HWIC-2T module into an open expansion slot on each router.
- Toggle the Power Switch back to ON.
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- 2.Connect Routers with Serial Cables:Physical Cabling.
- Select Connections (lightning bolt icon).
- Choose Serial DCE (red cable with clock symbol).
- Connect Router 1 (Serial0/1/0) to Router 2 (Serial0/1/0).
- Connect Router 2 (Serial0/1/1) to Router 3 (Serial0/1/0).
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- 3.Configure Router 1 Interfaces:CLI Configuration.
- Open the CLI tab on Router 1 and execute:
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```
Router> enable
Router# configure terminal
```

```
Router(config)# hostname Router1

! Configure LAN Interface
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit

! Configure WAN Serial Link to Router2
Router1(config)# interface Serial0/1/0
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
Router1(config-if)# clock rate 64000
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

- 4.Configure Router 2 Interfaces:CLI Configuration.
- Open the CLI tab on Router 2 and execute :

```
- Router> enable
Router# configure terminal
Router(config)# hostname Router2

! Configure WAN Link to Router1
Router2(config)# interface Serial0/1/0
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit

! Configure WAN Link to Router3
Router2(config)# interface Serial0/1/1
Router2(config-if)# ip address 10.0.0.5 255.255.255.252
Router2(config-if)# clock rate 64000
Router2(config-if)# no shutdown
Router2(config-if)# exit
```

- 5.Configure Router 3 Interfaces:CLI Configuration.
- Open the CLI tab on Router 3 and execute:

```
- Router> enable
Router# configure terminal
```

```

Router(config)# hostname Router3

! Configure LAN Interface
Router3(config)# interface GigabitEthernet0/0
Router3(config-if)# ip address 192.168.2.1 255.255.255.0
Router3(config-if)# no shutdown
Router3(config-if)# exit

! Configure WAN Link to Router2
Router3(config)# interface Serial0/1/0
Router3(config-if)# ip address 10.0.0.6 255.255.255.252
Router3(config-if)# no shutdown
Router3(config-if)# exit

```

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Task – 2 Now version 2 on three routers.

- RIP Configuration Steps :-
- 1.Enable RIP v2 on Router 1:Router 1 CLI.
- Open the CLI tab on Router 1 and advertise its connected networks (192.168.1.0 and 10.0.0.0):

```

- Router1> enable
Router1# configure terminal
Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# no auto-summary
Router1(config-router)# network 192.168.1.0
Router1(config-router)# network 10.0.0.0
Router1(config-router)# exit

```

- 2.Enable RIP v2 on Router 2:Router 2 CLI.
- Open the CLI tab on Router 2 and advertise both connected serial transit networks (10.0.0.0 and 10.0.0.4):

```

- Router2> enable
Router2# configure terminal
Router2(config)# router rip

```

```
Router2(config-router)# version 2
Router2(config-router)# no auto-summary
Router2(config-router)# network 10.0.0.0
Router2(config-router)# exit
```

- 3.Enable RIP v2 on Router 3:Router 3 CLI.
- Open the CLI tab on Router 3 and advertise its connected networks (192.168.2.0 and 10.0.0.0):

```
Router3> enable
Router3# configure terminal
Router3(config)# router rip
Router3(config-router)# version 2
Router3(config-router)# no auto-summary
Router3(config-router)# network 192.168.2.0
Router3(config-router)# network 10.0.0.0
Router3(config-router)# exit
```

- Execute show ip route on each router to confirm that all remote networks are dynamically learned (denoted by the code R).

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- 1. Router 1 Routing Table Output

- Router1# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

10.0.0.0/8 is subnetted, 2 subnets

```
C   10.0.0.0 is directly connected, Serial0/1/0
R   10.0.0.4 [120/1] via 10.0.0.2, 00:00:14, Serial0/1/0
C   192.168.1.0/24 is directly connected, GigabitEthernet0/0
R   192.168.2.0/24 [120/2] via 10.0.0.2, 00:00:14, Serial0/1/0
```

- Success Verification: Router 1 learns about Router 3's LAN (192.168.2.0/24) with a metric (hop count) of 2 via 10.0.0.2. Run ping 192.168.2.1 to confirm connectivity.

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- 2. Router 2 Routing Table Output

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- Router2# show ip route

10.0.0.0/8 is subnetted, 2 subnets

C 10.0.0.0 is directly connected, Serial0/1/0

C 10.0.0.4 is directly connected, Serial0/1/1

R 192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:08, Serial0/1/0

R 192.168.2.0/24 [120/1] via 10.0.0.6, 00:00:12, Serial0/1/1

- Success Verification: Router 2 learns both LAN subnets (192.168.1.0/24 and 192.168.2.0/24) with a metric of 1.

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- 3. Router 3 Routing Table Output

- Router3# show ip route

10.0.0.0/8 is subnetted, 2 subnets

R 10.0.0.0 [120/1] via 10.0.0.5, 00:00:18, Serial0/1/0

C 10.0.0.4 is directly connected, Serial0/1/0

R 192.168.1.0/24 [120/2] via 10.0.0.5, 00:00:18, Serial0/1/0

C 192.168.2.0/24 is directly connected, GigabitEthernet0/0

- Success Verification: Router 3 learns Router 1's LAN (192.168.1.0/24) with a metric of 2 via 10.0.0.5. Run ping 192.168.1.1 to confirm connectivity.

Task – 3 Adding the extra router and changing hop counter.

- Topology Transformation Overview :- Inserting Router 4 between Router 1 and Router 2 adds an extra hop to all traffic moving between Router 1 and the rest of the network.

- New Subnet Scheme :-

- Router 1 ↔ Router 4: 10.0.0.0/30

- Router 1 (S0/1/0): 10.0.0.1

- Router 4 (S0/1/0): 10.0.0.2

- Router 4 ↔ Router 2: 10.0.0.8/30 (New Transit Subnet)

- Router 4 (S0/1/1): 10.0.0.9

- Router 2 (S0/1/0): 10.0.0.10

- Router 2 ↔ Router 3: 10.0.0.4/30 (Unchanged)
- Configuration Steps :-
 - Add Router 4 & Rewire Serial Links: Cable Insertion.
 - Drag a new 2911 or 1841 Router into the workspace and name it Router 4.
 - Install an HWIC-2T module into Router 4 (remember to power off first).
 - Delete the existing serial cable between Router 1 and Router 2.
 - Connect Router 1 (S0/1/0) to Router 4 (S0/1/0).
 - Connect Router 4 (S0/1/1) to Router 2 (S0/1/0).
 - Configure Interfaces & RIP v2 on Router 4: CLI Setup. Open the CLI tab on Router 4 and run:


```
Router> enable
Router# configure terminal
Router(config)# hostname Router4

! Configure Serial link facing Router 1
Router4(config)# interface Serial0/1/0
Router4(config-if)# ip address 10.0.0.2 255.255.255.252
Router4(config-if)# no shutdown
Router4(config-if)# exit

! Configure Serial link facing Router 2
Router4(config)# interface Serial0/1/1
Router4(config-if)# ip address 10.0.0.9 255.255.255.252
Router4(config-if)# clock rate 64000
Router4(config-if)# no shutdown
Router4(config-if)# exit

! Enable RIP Version 2
Router4(config)# router rip
Router4(config-router)# version 2
Router4(config-router)# no auto-summary
Router4(config-router)# network 10.0.0.0
Router4(config-router)# exit
```
 - Update Router 2 Serial Interface IP: IP Update. On Router 2, update its Serial0/1/0 interface to match the new 10.0.0.8/30 subnet:

- Router2> enable
Router2# configure terminal
Router2(config)# interface Serial0/1/0
Router2(config-if)# ip address 10.0.0.10 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit
- Observing Hop Count Metric Changes
- After letting RIP converge (~30 seconds), inspect the routing tables using show ip route.
- 1. Router 3 Routing Table Output (show ip route) :- Check how Router 3 now reaches Router 1's LAN (192.168.1.0/24):
- Router3# show ip route

Gateway of last resort is not set

```

10.0.0.0/8 is subnetted, 3 subnets
R   10.0.0.0 [120/2] via 10.0.0.5, 00:00:12, Serial0/1/0
C   10.0.0.4 is directly connected, Serial0/1/0
R   10.0.0.8 [120/1] via 10.0.0.5, 00:00:12, Serial0/1/0
R  192.168.1.0/24 [120/3] via 10.0.0.5, 00:00:12, Serial0/1/0
C  192.168.2.0/24 is directly connected, GigabitEthernet0/0

```

- 2. Router 1 Routing Table Output (show ip route)
- Check how Router 1 now reaches Router 3's LAN (192.168.2.0/24):
- Router1# show ip route

Gateway of last resort is not set

```

10.0.0.0/8 is subnetted, 3 subnets
C   10.0.0.0 is directly connected, Serial0/1/0
R   10.0.0.4 [120/2] via 10.0.0.2, 00:00:15, Serial0/1/0
R   10.0.0.8 [120/1] via 10.0.0.2, 00:00:15, Serial0/1/0

```

C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
R 192.168.2.0/24 [120/3] via 10.0.0.2, 00:00:15, Serial0/1/0

- Metric Comparison Summary

Target Subnet	Initial Hop Count Metric (3 Routers)	Updated Hop Count Metric (4 Routers)	Path Breakdown (Traversed Routers)
192.168.2.0/24 (as seen from R1)	[120/2]	[120/3]	R1 → R4 → R2 → R3 (3 Hops)
192.168.1.0/24 (as seen from R3)	[120/2]	[120/3]	R3 → R2 → R4 → R1 (3 Hops)
10.0.0.4/30 (as seen from R1)	[120/1]	[120/2]	R1 → R4 → R2 (2 Hops)

- RIP increments the hop count metric by 1 for every router standard interface a packet must cross. Adding Router 4 increased the path cost to remote LANs from 2 hops to 3 hops ([120/3]).

Task – 4 Analyzing the RIP router updates.

- Step-by-Step Packet Capture Guide
- 1.Enter Simulation Mode:Switch Workspace.
- Look at the bottom-right corner of Cisco Packet Tracer.
- Click the Simulation Mode tab (or press Shift + S).
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- 2.Filter for RIP Packets Only:Protocol Filter.
- In the Simulation Panel, click the Show All/None button to uncheck all filters.
- Click Edit Filters.
- Under the MISC tab, check the box for RIP.
- Close the filter window. You should now see RIP listed under *Event List Filters*.
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- 3. Generate and Capture Update Packets: Capture Traffic.
- Click the Play / Capture Forward button (or hit Auto Capture / Play) to advance time.
- Since RIP broadcasts updates every 30 seconds, envelope icons (representing RIP updates) will appear on the router interfaces and travel to neighboring routers.
- Click the Stop button once an envelope reaches its neighboring router.
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- 4. Open and Inspect the RIP Packet (PDU): Packet Inspection.
- In the *Event List* on the right, click on any colored envelope icon corresponding to a RIP event.
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- The PDU Information window will open.
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- What to Analyze in the RIP PDU Information Window
- When you click on an event, inspect the following fields across the OSI layers:
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- 1. Inbound / Outbound PDU Details (Headers)

Layer	Protocol Field	Observed Value	Description / Significance
Layer 4	UDP Source Port	520	RIP uses UDP port 520 for sending updates.
Layer 4	UDP Destination Port	520	RIP listens on UDP port 520.
Layer 3	Source IP	Interface IP (e.g., 10.0.0.1)	The IP address of the router transmitting the update.
Layer 3	Destination IP	224.0.0.9	RIPv2 uses this dedicated Multicast group address to broadcast updates to all neighboring RIPv2 routers. (<i>Note: RIPv1 uses broadcast 255.255.255.255</i>).

- 2. In PDU Details Tab (RIP Header & Payload)
- Scroll down to the RIP Header section in the *In PDU Details* or *Out PDU Details* tab to view the actual advertised routes:.
- Version: 2

- Address Family Identifier: 2
- Advertised Route Entries:
 - IP Address: The network ID being advertised (e.g., 192.168.1.0).
 - Subnet Mask: 255.255.255.0 (*RIPv2 includes subnet masks, enabling VLSM*).
 - Next Hop: 0.0.0.0
 - Metric: 1

Packet Parameter	Captured Field Value
Transport Layer	UDP (Port 520)
Destination Address	224.0.0.9 (Multicast)
Update Interval	Every ~30 seconds
Payload Data	Subnet ID, Mask, Next-Hop IP, and Hop Count Metric

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